Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 1 ~ BookStream #002.0

BookStream | Parr, Pezzulo, Friston ~ Chapter 1 ~ BookStream #002.01 | 20:50

This section is going to be the chapter one and overview of the active inference textbook.

So shown here are the table of contents.

There are 10 chapters and three appendices in this book,

and they're separated into part one with chapters one through five and part two,

chapters six through 10.

Ali, do you want to give any overview thoughts on the organization or structure of the book?

Okay, so can you hear me now?

Sorry.

Yes.

Okay.

So part one is basically more aligned in terms of the theoretical construction of the active inference,

and it provides some conceptual and theoretical tools to be utilized in the second part,

because for the second part, we mostly deal with the practical side of active inference,

how to actually implement active inference to model some of the systems of interest.

So that's basically the main justification behind dividing up these chapters into two.

Great.

All right.

We're going to pass over the short preface written by Carl Fristen, but it's worth a read.

It's only several pages long.

So let's go to chapter one overview.

So do you have any opening thoughts on chapter one or where does one even begin with such a book?

Sure.

Yeah.

For chapter one, I think probably the most central theme of chapter one is about the two different approaches to active inference,

namely the high road and the low road.

And I think it provides a really nice perspective to view a kind of a whole picture and how the different components of active inference fit together.

So, but also I highly recommend chapter 13 from this book, Andy Clark and his critics written by chapter 13,

I think is entitled Beyond the Desert Landscape written by Carl Fristen,

in which he provides a more elaborate and a bit philosophical reasoning behind what exactly are low road and high road.

And what are the aims of each approach and the distinction between those two.

So for chapter one, we begin obviously with an introductory paragraph,

which sets the stage for the, I mean, all the components that would be discussed in this chapter.

So, but a very important section, I believe, is a section two here,
which is how do living organisms persist and act adaptively,
which is basically the main question active inference is trying to address.
So if we want to describe active inference in, I mean,
minimalistically with only one sentence or two,
probably we can say something to the effect that it's a theory that address this very question here.
So why they've opened up this chapter with important question.
And then action three, which is again a continuation of section one,
but a bit more specifically about active inference
and how active inference is addressing the question provided in the section two.
So basically again, active inference is a modeling framework for modeling the behaviors from first principles.
And what is meant by first principles will be elaborated upon in the following chapters,
following sections, and also obviously in the following chapters.
But very briefly, it's a kind of modeling framework that tries to somehow use variational principles,
such as free energy principle,
to construct modeling and mathematical tools
that will enable us to describe dynamical systems of living and even non-living agents
in terms of those variational principles
and how those agents or those systems are coupled with their environment.
So that's basically the unificatory principle that is at the heart of this active inference approach
because, yes, exactly here,
it's a kind of description of this action perception loop
that is at the heart of active inference or FEP framework.
framework.
And then we go to chapter, sorry, section 1.4,
which is again a description of the structure of the book
and why those two parts are divided as such.
So as I said earlier,
first part is more concerned with the theoretical underpinnings of active inference
and the second part is more about the practical side of it,
as we see in the subsection headings here.
So that's, yeah.
And then a very important diagram
which will provide very helpful
as we whole book is provided in figure 1.2,
which kind of ties all the different strands
of various related theories,
such as predictive coding, predictive processing,

Bayesian brain hypothesis, and so on,
into a kind of unified and integrated framework
by using those two trajectories of constructing active inference,
namely the high road and the low road.
So it literally here is represented as the high road
and the low road in this diagram.
So very briefly,
high road is a kind of top-down approach to active inference.
It begins with the question,
how things should act if they exist,
if they persist through time, sorry.
In other words,
it kind of sets the stage
for developing the concept of Markov blankets.
and then obviously by setting that also surprise minimization,
it attempts to elaborate that road
by adding other aspects of modeling,
other aspects of theory,
such as predictive processing,
self-evidencing, autopoiesis, and so on,
and then reach an integrated version of active inference,
which is basically an integration
between perception and action.
So I believe the main thing
that sets active inference apart
from all the other related theories,
such as predictive processing and predictive coding,
is the very deep integration of action and perception.
So action is not just some afterthought
or some additional component
that needs to be accounted for.
It is deeply woven into the fabric of the theory.
So as we'll see,
especially in chapters two and three,
even the mathematical formalism
for perception and action
are interestingly similar to each other
and even symmetric.

And it shows that perception and action
are actually not separate and distinct,
totally distinct entities and concepts.
And active inference definitely provides
a very promising framework
to integrate them into a holistic framework.
And another approach to construct that same theory
is to begin with the probabilistic description
of Bayes' theorem
and try to build up the formalism of active inference
using Bayes' inference theory,
Bayes' theorem and Bayes' statistical inference.
So again, we here see some very,
I mean, apparently different and distinct strands,
such as predictive coding,
Bayesian brain hypothesis and so on.
But as we see,
all of those different strands
can be united and tied together
within this low-road approach to active inference.
So I guess that's...
I'll just give one short thought.
Thanks.
These are great.
Yeah, just one short thought.
Thank you, Ali, for these great summaries.
The low-road and the high-road
are going to be explored in a lot more detail
in the coming chapters two and three.
We can think of the high-road as being a Y
because it describes how persistent
or remeasurable systems exist.
Now, they might be using Bayes' theorem
or they might be using
any other kind of mechanism internally,
but the high-road is describing the Y of existence.
From the low-road,
the bottom-up approach is kind of like a how.

You could use Bayes' theorem
to describe persistent autopoetic entities,
or you could use Bayes' theorem
for other purposes too.
So where these two roads intersect
is active inference,
and we have a lot more discussion coming
and, of course, unpacked in the textbook groups
about how these really map on
to bottom-up and top-down causation
and the plurality of Ys
that we approach in different settings.
Good to continue?
Yes.
Sure.
Some of the coming sections of this chapter,
continuing from page eight,
are going to describe what are happening
in the next chapters.
So chapter two is going to set out
the low-road perspective.
Chapter three is going to describe the high road.
Chapter four,
we will unpack active inference more formally.
That's one of the math-heavy chapters.
Chapter five,
we'll move from formal treatments
to biological implications of active inference
with a special focus on mammalian neurophysiology.
Chapter five sets out the process theory
associated with active inference
and gives a lot of hands-on and empirical examples.
And that's a lot of the first part of the book.
They're then going to summarize
some of the key points
and distillations from that theory-heavy first section.
One of them is that,
as Ali mentioned,

perception and action are complementary processes
and ways that fulfill the same imperative,
free energy minimization.
We sometimes summarize that by saying that
in the pursuit of the minimization of free energy,
agents can either change their mind
or change the world.
Change your mind
is associated with perception and learning,
and those are distinguished later.
And changing the world is associated with action.
There's some discussion about action selection
and optimal policy selection.
And all cognitive operations in active inference
are conceptualized as inference over generative models.
So the true kernel,
the cognitive kernel of active inference
is going to be the generative model.
Want to pick up from here
or give any thoughts on generative models?
I then carry on.
Thanks, Ali.
Just one point is,
there is a common misconception
about generative models,
which is that generative models
only refers to the inner states of the system
and also its markup blanket.
But actually, generative model
kind of encompass all the states of interest
of the situation we're trying to model.
So it also encodes the probabilistic information
about the external states as well.
So generative model can be thought of
as kind of the whole state of the system.
And by system here,
I mean the coupled dynamics
coupled dynamics between the internal and external states.

So that might be helpful to keep in mind.
Great.
One other distinction that comes up
and is used a lot nowadays,
although it's not always used in prior literature,
is the distinction between the generative model
and the generative process.
We're going to come back to it,
but usually the generative model
is being used to describe
our statistical model of the agent of interest.
And the generative process
is the process that generates observations
that are then passed
to the generative model as observations.
However, because generative models
can be generative processes for each other,
these two are not necessarily distinct in principle.
So in different simulations
and in different specific cases,
generative models can also be generative processes
for each other.
That leads us towards talking about
ecosystems of shared intelligence
and interactions amongst generative models.
But when we're talking about generative model,
it's like the organism
or the system of interest that we're focused on.
There are some more themes
that arise from the first half of the book.
Now we're on page 11.
Action is quintessentially goal-directed and purposive.
We talk a lot about how preference
plays a role in shaping action selection
in active inference,
and the goal-directed nature of active inference
will be unpacked in chapters two and three.
They discuss how various constructs

of active inference
have plausible biological analogs in the brain.
Once one has defined a specific generative model
for a problem at hand,
one can move from active inference
as a normative theory
to active inference as a process theory,
which makes specific empirical predictions.
And there's many, many interesting
philosophy of science type discussions
that we have about
explain, predict, design, control.
And then we get to section 1.4.2,
giving overview on part two active inference in practice.
So I'll just continue on.
Chapter six is going to introduce a recipe
for building active inference models.
So this is where the step-by-step comes into play.
It's going to give a sequence of steps
and approaches that you can take
to go from understanding the structural aspects
of a system or phenomena of interest
and build your way towards
having an active inference model
that can then be implemented in code.
And we have various examples
and we're there in the textbook group
to work with everyone
who wants to take this journey.
But chapter six is going to describe
that process of building
the active inference generative model.
So that's a recipe chapter.
And there's a lot more
that goes into the restaurant
also as we kind of have fun exploring.
Chapter seven and eight are like a pair
and they describe two different broad families

of generative models.
Now, these two families of generative models,
they work well,
they play well with each other,
but they are very educationally relevant
to also consider separately.
Chapter seven is going to focus
on the discrete time generative model.
And chapter eight is going to focus
on the continuous time generative models.
And again, these discrete
and continuous time models
can be nested and interoperable.
However, they do describe
some very important different situations.
And so chapter seven and eight
have a lot of technical information
on the discrete and continuous time
formulations of active inference
and also about how they integrate together.
Chapter nine is illustrations
of how active inference models
can be used to analyze data
from behavioral experiments.
And so this kind of goes beyond
chapter six's recipe
and it talks about some of the process
of making a behavioral experiment
and then using the outcomes
of that behavioral experiment,
the data from that experiment
and using that to parameterize
and do statistics
with the active inference generative model.
And chapter 10 brings it all together.
Chapter 10 is a lot like chapter one
and we've even joked
of reading the book backwards.

That's how great chapter 10 is.
So for those who are reading chapter one,
consider reading chapter 10 next
and then picking back up with chapter two
because chapter 10 is going to give
some judgments
and also sketch some exciting future directions
for where active inference is
in relationship to other fields
and where it's going.
I also second that.
Yeah.
The second part of the book
illustrates a broad variety
of models of biological
and cognitive phenomena.
So it's an application-oriented
second half of the book
and we have a lot of accessory code
and approaches and notebooks.
So one may not find
that this is the one-stop shop
that they were looking for.
However, for those who dive in,
there's absolutely more than enough
to scaffold their learning journey.
And then section 1.5 is the summary.
The chapter introduces
the active inference approach
to explain biological problems
from a normative perspective
and previews some implications
of this perspective
that will be unpacked
in later chapters.
It describes the structural outline
of the book,
chapters one through five

being theory,
chapters six through ten
being practice,
and then it signals
that the next two chapters
are going to develop
the low and the high road perspectives
and altogether,
that is chapter one.
Any other thoughts on chapter one?
Not specifically,
but I think it provides
a really good context
to the rest of the book
in terms of
how all the
different elements
of active inference
comes into play
in addressing
different problems,
different problems.
Awesome.
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